HW1 Bonus: MNIST Exploration

Real-World Dimensionality Reduction

Course: Machine Learning for Single-Cell and Spatial Biology (Prof. Uthsav Chitra, JHU)
Unit 1 — Linear Dimensionality Reduction (PCA, NMF, GLM-PCA) | Prepared by Sai Dharmasena

Overview

This is an open-ended, exploratory exercise using a real-world dataset: a subsection of MNIST handwritten digits. Unlike the 5 simulated datasets (which each have a hidden artifact to spot), this exercise has **no hidden artifact**. Students are given real data and asked to apply whatever skills they've learned in Unit 1 — PCA, NMF, normalization, visualization — to explore and understand the data.

The goal is for students to:

- Practice dimensionality reduction on real (not simulated) data
- Discover that the same concepts from class apply to non-biological data
- Build intuition for what PCA and NMF "see" when applied to images
- Get creative — there's no single right answer

The Dataset

File: `data/mnist_subsection_expression.csv` (1500 images × 784 pixels)
Metadata: `data/mnist_subsection_metadata.csv` (digit label per image)

- 3 digits: **0, 1, 7** (500 images each, shuffled)
- Each image is 28×28 = 784 pixels, values 0–255
- 82.7% of pixels are zero (sparse, like scRNA-seq!)
- Pixel columns named `Pixel_000` through `Pixel_783` (row-major order: pixel = row×28 + col)

> **Analogy to single-cell data:** Cell = Image, Gene = Pixel, Gene expression count = Pixel intensity, Cell type = Digit identity, Library size = Total brightness, Highly variable genes = High-variance pixels.

Suggested Exploration Directions

Students can take this in any direction. Here are some ideas:

PCA

- Run PCA on raw pixel values. Do the 3 digits separate in PC space?
- Look at the **PC1 loading as an image** — what does the direction of maximum variance "look like"? Is it interpretable?
- How many PCs are needed to explain 90% of the variance? Plot the variance explained curve.
- Try PCA with and without z-scoring pixels. Does it change the result? Why?
- Visualize PC1 vs PC2, PC2 vs PC3, etc. — different PCs capture different structure.

NMF

- Run NMF on the raw pixel values (they're non-negative — perfect for NMF).
- Look at the **NMF components as images** — do they correspond to "parts" of digits (strokes, loops, lines)?
- How many components do you need? Use the elbow method on the reconstruction error.
- Compare NMF components to PCA components — which are more interpretable? Why?
- Can you reconstruct a digit from just a few NMF components? How does the reconstruction quality compare to PCA?

Normalization / Preprocessing

- Some images are brighter than others (more ink). Is this analogous to library size in scRNA-seq?
- Color your PCA plot by total brightness. Does PC1 correlate with brightness?
- Try normalizing each image to the same total intensity. Does this change the PCA?
- What happens if you binarize the images (threshold at 128)? Does PCA still work?

Creative / Open

- Can you use PCA or NMF to classify digits? (e.g., train a simple classifier on the top 10 PCs)
- What happens if you add noise to the images? Can PCA denoise them?
- Try PCA on just one digit — what does the within-digit variation look like?
- Compare the structure of digit 1 (simple, mostly vertical line) vs digit 0 (loop) in PC space.

Deliverables

This is exploratory — students should submit:

- **A notebook** showing their exploration (code + plots + commentary)
- **3–5 figures** with captions explaining what they show
- **A short reflection** (1 paragraph): What did they learn? What surprised them? How does MNIST relate to single-cell analysis?

Grading (Exploratory Rubric)

Criterion	Points
Applies at least 2 different methods (PCA, NMF, etc.)	3
Produces clear, labeled visualizations	3
Provides thoughtful interpretation of results	3
Makes connections to single-cell concepts from class	3
Reflection demonstrates genuine exploration	3
Total	15

Example Figures (for instructor reference)

Four example figures are included showing what students might discover:

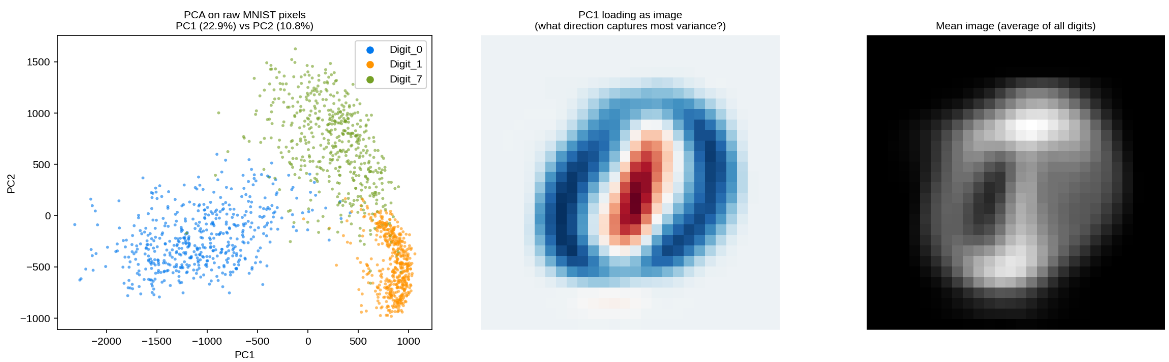

Figure 1: PCA on raw MNIST pixels. Left: PC1 vs PC2 colored by digit. Center: PC1 loading as a 28x28 image. Right: Mean image.

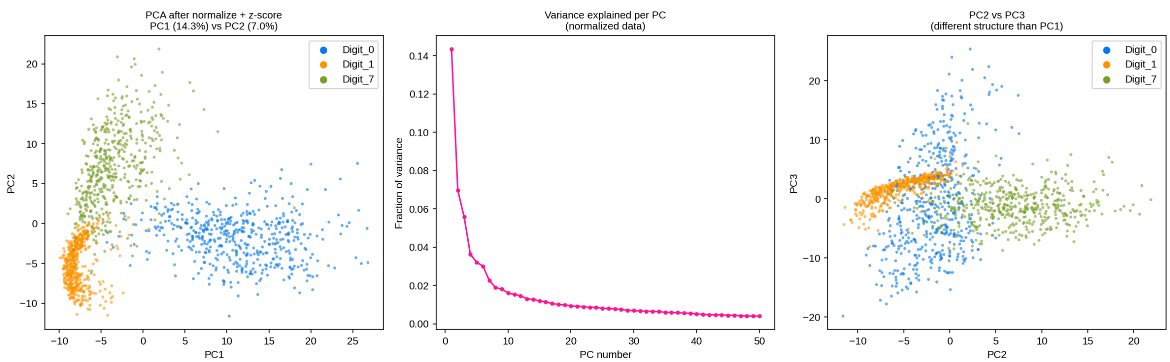

Figure 2: PCA after normalization. Left: cleaner digit separation. Center: variance explained curve. Right: PC2 vs PC3.

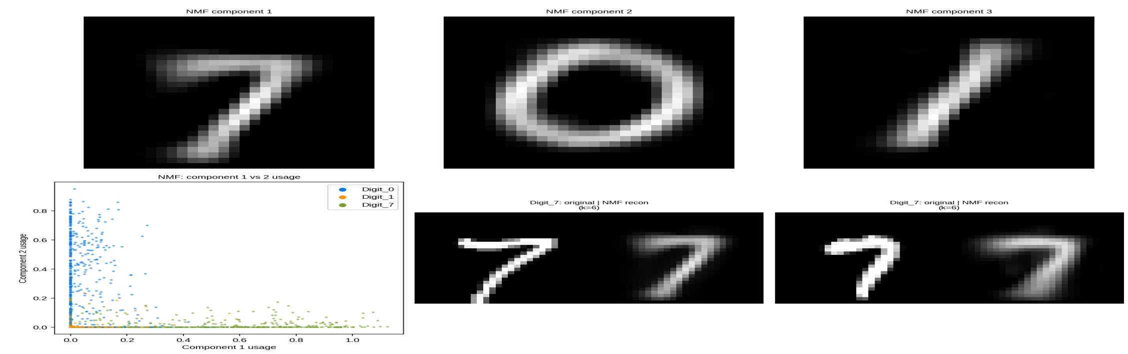

Figure 3: NMF components as images (top) — they look like "parts" of digits. Bottom: NMF usage scatter and original vs reconstructed images.

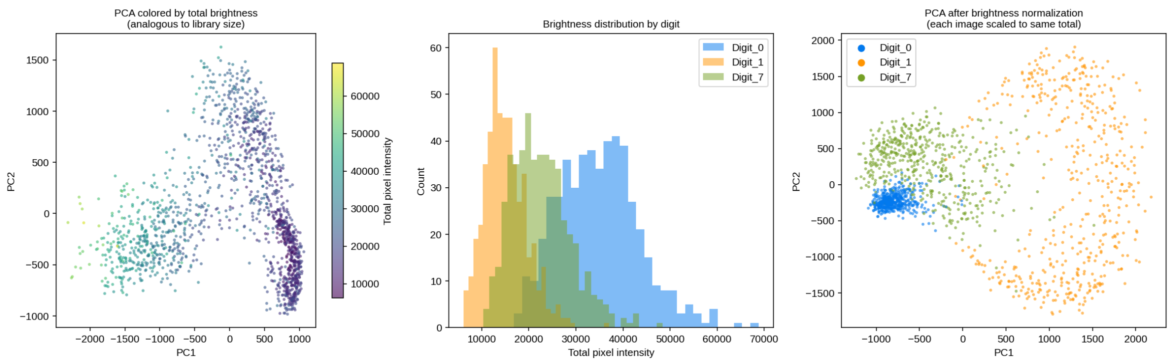

Figure 4: Brightness as "library size." Left: PCA colored by total brightness. Center: brightness distribution by digit. Right: PCA after brightness normalization.

Notes

- **No hidden artifact** — this is pure exploration. The brightness variation is natural, not injected.
- **The scRNA-seq analogy is the key teaching moment.** Students should recognize that "total pixel intensity" in MNIST is the same concept as "library size" in scRNA-seq, and that PCA on raw data can be confounded by it in both cases.
- **NMF is particularly intuitive on MNIST** because the components are directly visualizable as image parts — this makes the "parts-based, additive" interpretation from class_04 concrete in a way that gene programs aren't.
- **Students can use an AI agent** to help download, preprocess, and visualize the data — this aligns with the course's goal of using agents for data analysis.
- **The dataset is small (1500 x 784)** so everything runs in seconds.